House Price Prediction using NumPy: Mini Report

1. Objective

The objective of this project is to predict the price of a house from its size using simple linear regression, built with NumPy.

2. Dataset

- **Feature (input):** house size in square feet.
- **Target (output):** house price in lakhs of rupees.
- The dataset has 10 houses. Sizes range from 500 to 2800 sq ft, and prices range from 15 to 84 lakh.
- The average size is 1630 sq ft and the average price is 48.9 lakh. The median price is 49.5 lakh.
- **Training data:** the first 8 houses. **Test data:** the last 2 houses (2500 and 2800 sq ft).

3. Method

First, I found the mean of the training sizes (1375) and the mean of the training prices (41.25). Then I calculated the slope and the intercept using the closed-form formulas of simple linear regression:

- Slope: w = numerator / denominator = 76050 / 2535000 = **0.03**
- Intercept: b = y_mean - w x x_mean = 41.25 - 0.03 x 1375 = **0**

So the model is: **Predicted Price = 0.03 x Size**. This means every extra square foot adds 0.03 lakh to the price.

How the model works: the model looks at the training houses and finds the straight line that fits them best. It then uses this line to predict the price of any new house size. I wrote a function, `predict(size)`, which returns `w x size + b`. For a 1300 sq ft house, it predicts **39 lakh**. NumPy makes this easy because it can do calculations on a full array at once, without loops.

4. Results

- **Test predictions:** the model predicted 75 lakh for the 2500 sq ft house and 84 lakh for the 2800 sq ft house. These are the same as the actual prices.
- **Errors:** 0 and 0.
- **MAE (Mean Absolute Error) = 0.0 lakh**
- **MSE (Mean Squared Error) = 0.0**
- MAE is easier to understand because it is in the same unit (lakh) as the price. MSE is in lakh squared, which is hard to imagine.

Challenge A: Unusual house. I added my own unusual house: 1600 sq ft priced at 250 lakh. The mean price changed from 48.9 to 67.18, but the median moved only from 49.5 to 54.0. The slope changed from 0.03 to 0.0289, and the prediction for 1300 sq ft changed from 39 to 57.71 lakh. This shows that one strange house can pull the line towards itself and make the predictions for normal houses wrong.

Challenge C: Baseline comparison. When I always guessed the average training price (41.25 lakh), the MAE was **38.25 lakh**. My linear regression model had an MAE of **0.0 lakh**, so the model is much better than the baseline.

5. Reflection

What worked: the linear regression model gave perfect predictions because this dataset is clean and almost perfectly linear. NumPy made the calculations short and simple.

What I learned: I learned the full workflow of a small machine-learning project: storing data in arrays, exploring it with statistics, splitting it into training and test sets, training a model, evaluating it with MAE and MSE, and making the program interactive with input and error handling.

Two limitations:

1. The dataset is very small (10 houses), and only 2 houses were used for testing. A zero error on 2 test examples does not prove that the model will work well in the real world.
2. The model uses only house size. Real house prices also depend on location, age, number of bedrooms, amenities and market conditions.